

Graphical Abstract

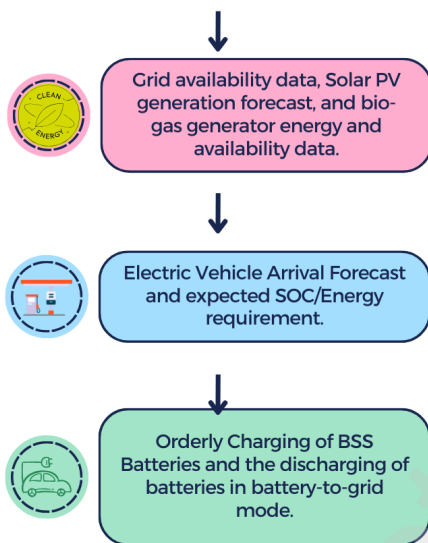
Multi-Objective Optimization of Battery Swapping Station to Power Up Mobile and Stationary Loads

Muhammad Shuzub Gull, Muhammad Khalid, Naveed Arshad

MULTI-OBJECTIVE OPTIMIZATION OF BATTERY SWAPPING STATION TO POWER-UP MOBILE AND STATIONARY LOADS

The growing trends of Electric Vehicles requires sophisticated and quick recharging infrastructure. Battery Swapping Station provides fully charged batteries to Electric Vehicles on demand in exchange of discharged batteries. Therefore, the swapping station infrastructure must be explored for increased profitability.

Objective



Methodology & Evaluation

Model	Process	Result
Generation Optimization	Generation mix comprising weak grid, solar PV, and biogas generation to provide the minimum cost of battery charging and reduce the cost.	The cost of battery charging is reduced by 34% as compared to the base case with weak connected grid.
BSS Optimization	Orderly charging of discharged batteries to allow BSS to operate in grid-to-battery mode and increase revenue.	The revenue is increased by 18% as compared to the base case with weak connected grid.
Multi-Objective Optimization	Combination of Generation and BSS Optimization Modules to reduce cost and increase revenue at the same time.	- The daily net profit of BSS is increased by 20 times as compared to the base case with weak connected grid. - When BSS operates on battery-to-grid mode, the profit is increased by 8%.

Highlights

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- Accurate EV arrival and energy forecasts crucial for station planning.
- Use mixed resources like weak grid, solar PV, and bio-gas for energy
- Idle batteries operate in battery-to-grid mode in underserved markets.
- Model boosts revenue and cuts costs at swapping stations.
- Model simulates station modes to boost profit in the EV market.

Multi-Objective Optimization of Battery Swapping Station to Power Up Mobile and Stationary Loads^{*,**}

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ABSTRACT

The increasing emphasis on clean emission-free transportation has stressed upon the high adoption and penetration of electric vehicles worldwide. However, challenges like range anxiety and long charging time hinder the widespread acceptance. Among various solution the usage of battery swapping station seems more promising as it provide quick battery refueling within a very short time period. The battery swapping station's progress is limited due to the associated investment and operational cost which needs to be addressed to ensure the global acceptance. In this paper, an optimal battery swapping station operation is proposed based on a multi-objective optimization which combines the generation mix of grid, solar PV, and biogas generation along with the battery arrival using mixed integer programming and orderly charging of discharged batteries to allow the swapping station to operate in battery-to-grid mode using time-dependent dynamic programming. The former reduced the cost of charging while the later increases the swapping station revenue. The combined multi-objective optimization increases the daily net profit by almost 20 times as compared to the base case and by 8% in the optimal operation when the batteries are allowed to be discharged in battery-to-grid mode. The results show that the proposed strategy has potential to significantly boost the daily net profit of the swapping station by increasing revenue and cut cost and can contribute towards the acceptance of swapping station in the current electric vehicle market.

Nomenclature

η^{chg} Charging Efficiency

$\gamma^{allowed-dis}$ Percentage of batteries allowed to be discharged at a time

γ^{chg} Percentage of energy above which battery charging is initiated

γ^{dis} Percentage of energy beyond which battery discharging is not allowed

$\gamma^{max-dis-limit}$ Percentage of EV arrivals below which battery discharging is allowed

γ^{PCB} Percentage of energy above which battery swapping is allowed

bg_t^{avl} Binary variable to identify if biogas generator is available at time t

C Cost

C^{chg} Cost of charging

$C^{fix-cost-chg}$ Fix cost associated with battery charging cycles

$C^{fix-cost-dis}$ Fix cost associated with battery discharging cycles

$chg_{t,b}$ Binary variable to identify if battery b is being charged at time t

$dis_{t,b}$ Binary variable to identify if battery b is being discharged at time t

$E_{t,b}'^{chg}$ Change in energy of battery b charging at time t

$E_{t,b}'^{dis}$ Change in energy of battery b discharging at time t

E_t^{BSS} Energy of BSS battery b at time t

$FC_t^{fix-chg}$ Fix cost of battery charging cycles at t

$FC_t^{fix-dis}$ Fix cost of battery discharging cycles at t

g_t^{avl} Binary variable to identify if grid is available at time t

$N^{Battery}$ Battery capacity of BSS

$N^{chg-ports}$ Number of charging ports at the BSS

N_t^{chg} Number of batteries being fully charged at t

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In this work we optimize the battery swapping station operation and integrate it with the generation optimization module to boost the daily net profit.

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N_t^{dis}	Number of batteries being discharged till the threshold at t
P_t^{BSS2EV}	Price of per unit electricity offered by the BSS to EV in energy trade
P_t^{opt}	Optimal cost of electricity for battery charging at t
Q_b	Nominal capacity of battery b
R	Revenue
R^{dis}	Revenue from battery discharging
$R^{swap-energy-traded}$	Energy Trade Revenue from battery swapping
$R^{swap-fix-cost}$	Fixed Revenue from battery swapping
R^{swap}	Revenue from battery swapping
$swap_{t,b}$	Binary variable to identify if battery b is swapped at time t
BESS	Battery Energy Storage System
BSS	Battery Swapping Station
DB	Depleted Battery
EV	Electric Vehicle
FCB	Fully Charged Battery
FCBI	Fully Charged Battery Inventory
PCB	Partially Charged Battery
SOC	State of Charge
SOH	State of Health
T	Number of Time Period in 24 hours
T^*	Subset of T where the number of EV arrival is less than the discharging threshold

1. Introduction

With the widespread acceptance and penetration of Electric Vehicles (EV) in the transport sector a significant reduction is observed in the emission of green house gases (GHG) in the Earth's atmosphere and therefore, the carbon foot print has been reduced (Farzaneh and Jung (2023)). Given the reduced emission and competitive advantage EV have the potential to replace conventional fossil fuel vehicles. The United Nations Climate Change (UNCC) has established certain climate goals, by limiting the temperature rise by 1.5 °C and reducing GHG by 43% (Agreement (2015)). The completion of such goals is necessary for the survival and sustenance of climate and atmosphere and to achieve them the clean energy resources and clean emission-free vehicles must be introduced in the consumer market. But the required change is associated with potential challenges

which need to be addressed. Among the challenges associated with EV adoption is the range anxiety and long charging time which can be addressed through advanced charging methods, swapping stations, or hybrid models, etc.

One of the possible solutions to the EV range anxiety and long charging time is the use of efficient and fast Battery Charging Station (BCS), although the problem is mitigated but such fast charging results in accelerated battery degradation and thus they are not widely utilized (Tomaszewska, Chu, Feng, O'kane, Liu, Chen, Ji, Endler, Li, Liu et al. (2019)). Typical BCS have a three level charging infrastructure where the first level, i.e., L1, are slow chargers, second level, i.e., L2, are medium chargers, while third level, i.e., L3, are fast chargers taking no more than 45 minutes to recharge EV battery. Despite the quick recharging of EV at L3, the overall life is reduced along with the performance efficiency (Wu, Pang, Choy and Lam (2017)).

Another possible solution is to use a Battery Swapping Station (BSS) where the EV can quickly swap/replace its depleted battery (DB) with a fully charged battery (FCB) within a few minutes. Such an architecture can extend EV range by exchanging discharged batteries with charged ones Ghosh (2020). BSS architecture ensures rapid EV refueling while optimizing the battery charging schedule of received discharged batteries. However, this architecture incurs extra capital and operational cost due to the requirement of additional infrastructure, batteries along with the monitoring and control of battery stock, battery state-of-health (SOH), and battery state-of-charge (SOC). This hinders the success and adoption of BSS architecture in the current EV market.

To cater for the additional cost requirement of the BSS, the said architecture can be integrated with solar PV generation which enables clean charging energy and a cost-effective recharging solution. Such an optimization model can reduce the cost of electricity by orderly charging batteries resultant causing the peak shaving of the demand profile (Wang, Ma, Liu and Wang (2021)). The solar PV integrated BSS modeled by mixed-integer linear programming (MILP) shows that the model generates high profits compared to traditional grid connected BSS given the inherent uncertainties are under a controlled environment (Cheng and Zhang (2017)). Therefore, a time-series forecast of EV arrival time, energy demand, solar energy output and crucial to plan the next charging schedule of the BSS operation.

Moreover, to increase the profitability of the BSS, several different operations of operation have been opted such as market participation for frequency, ancillary service or even discharging batteries to grid in a day-ahead market. Such operational modes are stimulated by the grid price where battery inject energy to the grid when the price is high and charge batteries when the price is low have been observed to be economically profitable for the BSS operation (Sarker, Pandžić and Ortega-Vazquez (2013)). However, in developing countries most of the electricity markets are not operational and the power sector operation on traditional single buyer or monopoly models such market participation is not possible. The only solution in such a power system to

discharge batteries to the grid irrespective of the day-ahead market based on the price signal and battery energy storage component.

With the accurate knowledge of EV arrivals, energy requirement, supply and demand, and power system dynamics, the BSS operation can be optimized such that it can generate increased profit while successfully fulfilling the EV demand. Moreover, the exploration of multiple revenue stream for the BSS operation as battery-to-grid can enhance the revenue resulting in increased profit. Such optimization can boost the BSS profitability by enhancing resource utilization, and cost reduction to promote the adoption and penetration of the BSS architecture.

This paper proposed an innovative way of enhancing the BSS architecture profit in a developing country which utilizes the portable battery-based storage system in the country's power system. We introduce a novel BSS-based multi-objective optimization based on the generation optimization and BSS operational optimization strategy that leverages the grid availability, solar energy generation forecast, biogas energy generation for the generation mix operation. Since, the ongoing pilot project of the BSS is located in a rural area near a farmland where we have abundance of animal manure and organic waste to obtain biomass. This biomass is currently being used to extract cooking gas while the by-product can be used to generate electricity. Based on the project setup, the proposed strategy utilizes the biogas in generation mix for the generation optimization model which would reduce the cost of charging. Furthermore, the EV arrival forecast and energy demand is used to generate an orderly battery charging schedule to allow battery discharging to the grid in a BSS optimization model which would increase the BSS revenue. The proposed strategy integrates both models such that the former reduces the cost of charging while the later increase the BSS revenue, resulting in increased daily net profit. The aim is to upgrade the BSS to operate in a inter-sectoral environment working along side energy, food and transportation nexus (EFTN).

The rest of the paper is arranged as follows: The literature review in Section 2 presents the latest research in the area while identifying the gaps and areas of improvement. Section 3.1 defines the problem statement along with the description of data being used. Section 4 highlights the proposed methodology of the underlying architecture. Section 5 describes the mathematical formulation of the proposed methodology. Section 6 assigns the parameter settings and values to be taken in simulations. Section 7 demonstrates the simulation output and discuss results and implication. Section 8 provides the conclusion of the research study.

2. Literature Review

The integration of renewable energy resources such as Solar energy with the existing transmission and distribution network has been the highly beneficial and has potential to lead towards renewable energy-driven future for the power

sector (Rauf, Gull and Arshad (2020)). The inherent intermittency of renewable energy resources leads to supply-demand mismatch and fluctuations. These fluctuations are managed by using BSS in a battery-to-grid model where BSS provides services such as voltage support to the grid in addition to provide services to EV (Vallera, Nunes and Brito (2021)).

A multi-objective BSS optimization is used to orchestrate the battery charging schedule to minimize grid load fluctuations and maximizing the revenue using the generation mix of solar and wind which diminishes the fluctuations and increases economic parameters (Wang et al. (2021)). Similar concept uses the digital twin of microgrid to integrate second life BSS batteries, utilities, and grid along with Solar and Wind as renewable resources to strive for profit and efficient utilization of second life batteries (Marchesano, Guizzi, Vespoli and Ferruzzi (2023)).

A MILP formulation is used to optimize the BSS size such that the system is integrated with grid and solar PV generation. This capacity planning optimization problem uses Monte Carlo simulations to compare performance and generates effective utilization of BSS resources within the optimal capacity producing maximum profits (Ban, Guo, Yu and Shahidehpour (2019)). Another approach uses the self-consumption and self-sufficiency parameters along with F1 scoring to optimize the size of solar power BSS. The scheme shows that the charging schedules improve the load matching performance and the inherent simplicity implies that the model can be used as a benchmark (Fachrizal, Shepero, Åberg and Munkhammar (2022)).

Another approach uses the Solar PV generation to recharge batteries as the BSS keeping the grid as last resort. To improve the model's output and accuracy the proposed strategy uses the Long Short-Term Memory (LSTM) forecast as the model is highly dependent on the forecast for power scheduling from the grid and other renewable resources (Chawrasia, Hembram, Bose and Chanda (2023)). A smart grid minimize supply and demand gap by using demand response programs such as soft load shedding or brownout strategies to generate cost-effective electricity (Gull, Mehmood, Rauf, Khalid and Arshad (2022)).

The BSS architecture has potential to be integrated into various ancillary services such as battery-to-grid, battery-to-battery etc. which allows the operational flexibility and optimized charging schedules (Revankar and Kalkhambkar (2021)). Typically the operational optimization of the BSS batteries is carried out by the Battery Management System (BMS) which monitors and controls the voltage, current, temperature and safety factors such as thermal runaway and it is crucial for the optimal operation of the BSS (Gabbar, Othman and Abdussami (2021)).

A detailed research combines the operational scenarios of the BSS combined with the Solar PV integration in a distributed generation system. Several operational modes such as battery-to-grid, grid-to-battery, battery-to-battery, PV-to-battery, and PV-to-grid have been explored in terms of energy and financial transactions. The model considers

the quality of service and battery state-of-health as necessary constraint to determine a working business model on various battery capacities and solar PV production profiles (Erđinç (2021)).

In most BSS operations aggregators are used which are installed to maximize the profit on daily basis, as the aggregate the demand to match demand and supply and perform the best available energy transactions while balancing the demand and supply constraints. Using such a methodology a centralized battery charging infrastructure generates more profit than a BSS (Gupta, Kumar, Panigrahi and Al-Sumaiti (2020)). The negative impact of grid connected solar PV based BSS such as system fluctuations can be mitigated using the energy storage component of the BSS as backup support to the installed generation in a service availability and self-consumption model (Liu, Chen, Lu, Liu and Zhang (2015)). BSS generates surplus revenue by participating in electricity market using a dynamic model especially during high price periods as it traces the battery states during the participation (Yang, Yao, Kang and Zhu (2014)).

To increase the profits of the BSS linear programming is applied to reduce the cost of charging expense along with other day-to-day expenses using orderly charging and discharging schedule of EV taxis, the model also suggest the factors influencing the BSS operation based on the sensitivity analysis (Liang, Zhang, Xie and Liu (2017)). Similarly for public transport such as EV buses, the size optimization of the BSS depends on the net present value (NPV) which is determined based on bus specific route. Moreover, the BSS size correlates with the electricity sold by the grid (Zheng, Dong, Xu, Meng, Zhao and Qiu (2013)). A stochastic approach forecast the EV bus load demand for unknown EV Bus swap demand without reservation to efficiently schedule battery charging (Dai, Cai, Duan and Zhao (2014)). Furthermore, the utilization of BSS for private EV has been proven beneficial and has potential for widespread adoption.

To maximize the profit of BSS, the strategy uses the idle BSS batteries to participate in frequency regulation market which explores a new revenue stream of BSS resulting in increased profit (Zhang, Yao, Zeng, Yang, Zhao, Lai and Lai (2021)). The battery scheduling reduces the electricity cost while maximally utilizing the available Solar PV generation (Shalaby, Shaaban, Mokhtar, Zeineldin and El-Saadany (2021)). A few studies propose a business model on the optimal charging and bidding strategies based on the market dynamics, consumer and EV behavior, and data driven models for multi-charging scenarios for plug-in EVs to maximize the profits (Ahmed, Rehan, Basit, Tufail and Hong (2023b), Ahmed, Rehan, Basit, Tufail and Hong (2023a)).

To address the inherent uncertainties of solar PV generation and EV demand forecast of the BSS chance-constraint programming method is used which minimizes the cost of electricity purchased from the grid within a controlled probability of constraint failure. The results shows significant improvement in performance and reduces the cost of electricity (Liu, Zhang, Ge, Gu and Li (2019)). The

Alternative Method of Multiplier (ADMM) method is used to optimize both the microgrid and the BSS. The method targets cost reduction and congestion avoidance based on the depth of discharge and battery degradation (Esmaeili, Anvari-Moghaddam and Jadid (2019)). A cyber-physical model integrates the nanogrid with BSS architecture to increase the energy supply, reliability, and profitability using MILP formulation. The method uses battery energy storage of the BSS to minimize the investment cost of planned PV energy storage system (Ban, Shahidehpour, Yu and Li (2017)).

A ranking system is introduced to mitigate the congestion of EV taxis at the BSS which translates to efficient scheduling schemes (Wang, Ding, Huang, Wei, Liu and Stankovic (2017)). Within the battery charging station (BCS), the charging schedule optimizes the charging level to minimize the cost of electricity and fulfill demand however, the BSS architecture outperforms the BCS architecture in terms of flexibility and cost of services offered (Tan, Qu, Sun, Li and Tsang (2017)). While many studies consider the battery to be same and homogeneous, a close to real-world application optimize the BSS operation with inherent battery heterogeneity and generate cost effective charging schedules (Zhang, Cao, Peng, Ahmad and Xu (2020)).

The technical, economical, and environmental aspects of the EV charging stations have been explored powered by the hybrid generation resources which include the combination of Solar PV/Wind Turbine, Diesel Generation/Battery with types including lead-acid battery, flow-zinc-bromine battery, and lithium-ion battery. The assessment is made on the net present value (NPV) of the assets and the combination has been found efficiency (Muna and Kuo (2022)). The reduction of weighted sum of cost of charging can be achieved by using the Solar PV based BSS optimized using the Markov Decision Problem (Jin, Mao and Xu (2023)).

A dynamic BSS optimization model uses the Long Short-Term Memory (LSTM) based forecast joined with Rolling Horizon Optimization (RHO) to cater for 24 hour day operation leading into the next day. The model incorporates battery heterogeneity and facilitates the energy to the grid during peak load hours offering battery-to-grid, grid-to-battery, battery-to-battery and ancillary services with improved performance based on Root Means Square Error (RMSE) (Shalaby, Shaaban, Mokhtar, Zeineldin and El-Saadany (2022)). The preliminary analysis consider the BSS to supply excess energy to a nearby load on a farmland area and results show the cost of electricity offered to the load from the backup energy storage of the BSS is significantly less than the grid and also generates profit for the BSS (Gull and Arshad (2022)).

Extensive research has explored the BSS's technical, economical and financial aspects to develop business models using advanced Machine Learning (ML) algorithm to boost profit and reduce cost. Moreover, several methodologies have incorporated the hybrid renewable generation such as Solar PV and Wind to reduce the cost of battery charging at the BSS. Similarly, several studies have explore the revenue

streams such as battery-to-grid, and battery-to-battery to enhance the BSS profit. However, the use to biogas generation in the generation mix has not yet been explored. Since, our ongoing project has such a biogas generator available and can be used to provide the generation capacity, therefore, this area of research needs to be explored. This study introduces an innovative multi-objective optimization strategy which tries to reduce the cost of electricity required for battery charging by integration grid, solar PV, and biogas optimal generation, while trying to maximize the revenue by allowing the idle BSS batteries to discharged and sell excess energy to the grid based on price signal and number of EV arrivals. The next section outlines the problem statement and the dataset.

3. Problem Description

This section describes the problem statement in 3.1, and the data available and discuss the pre-processing of the dataset which is being used in the optimization in 3.2.

3.1. Problem Statement

The research problem caters to maximize the daily profit of a BSS by increasing the revenue of the BSS by utilizing the excess battery energy storage available and by reducing the cost of battery charging and auxiliary consumption using an optimal energy generation mix comprising of grid energy, solar energy, and biogas energy while considering the battery energy storage component of the BSS as a backup to the connected unreliable grid.

3.2. Data Description

The dataset comprise two sections, first section contains EV arrival data with associated state of charge (SOC). Second section contains the price of electricity offered by the grid along with the grid availability, i.e., grid availability identifies the period when the grid is able to feed the demand, the price of electricity and energy provided by the installed solar panels, and the price of electricity, energy offered and availability period of the connected biogas electricity generation plant over 24-hour time horizon. The obtained dataset is pre-processed and is provided to the proposed architecture which is bound to real-world constraints including demand and supply balance, battery energy storage limitations, generation mix and optimal operation of the integrated system.

The EV arrival data set consists of number of EV arrivals expected in the each time period $t = 1, 2, \dots, T$ along with the expected SOC of each EV arriving in the particular period. The EV arrival follows Poisson distribution such that the inter-arrival time is mutually exclusive with an arrival rate of λ [customers/second]. The probability that the BSS has k arrival at the rate of λ in the period t is given by 1:

$$P(A(t) = k) = \frac{(\lambda t)^k}{k!} \exp(-\lambda t) \quad (1)$$

Where $A(t)$ is the number of EV arrivals expected at time t for the Poisson distribution given by 2.

$$A(t) = \sum_{i=0}^{\infty} 1_{T_i \leq t} \quad (2)$$

Moreover, the service time of the BSS is given by μ [customers/second] such that the offered load is given by $\rho = \lambda/\mu$. Since, the service of battery swapping is considered constant for simplicity, therefore, the service time distribution is modeled as Deterministic, and the BSS has K number of charging ports available with a huge buffer to simplicity and FIFO as default service discipline. Based on the available characteristic the dataset can be denoted in Kendall's notation as $M/D/K$ system. The given random process has a specific structure where the future events depends on where the current observation stands and not where it has been in the past, i.e., memorylessness.

Furthermore, the EV SOC estimates follow normal distribution with mean μ_{SOC} and standard deviation σ_{SOC} whose probability density function (PDF) is given by 3.

$$f(x) = \frac{1}{\sigma_{SOC} \sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x - \mu_{SOC}}{\sigma_{SOC}}\right)^2\right) \quad (3)$$

The generation dataset consist of grid data, solar PV data, and biogas energy generation data. The grid data comprises the maximum grid capacity denoted by G_{max} , and the grid electricity price including peak and off-peak tariff along with the load management/load shedding schedule which identifies the periods when the grid is not available. The solar PV data comprises the expected/forecast solar PV energy output, and the offered electricity price. The solar PV power and energy output are given by 4 and 5 respectively.

$$P_{out} = P_{STC} * 1 + \alpha_{TC} * (T_{cell} - T_{STC}) \quad (4)$$

$$E_{out} = P_{out} * t \quad (5)$$

Where P_{out} and E_{out} are power and energy output of solar PV panels, P_{STC} is rated power output (in kW) at Standard Test Conditions (STC), α_{TC} is the temperature coefficient (TC) of power (in %/°C), T_{cell} is the actual temperature of solar cell (in °C), and T_{STC} is the STC value of temperature of solar cell (in °C). The levelized cost of electricity (LCOE) offered by the Solar PV generator embeds the 25-year lifespan of the installation in the operational cost of the model.

The biogas electricity data comprises the biogas generator capacity, the the maximum available energy and power output data, the offered electricity price and the biogas availability data which identifies the period when the biogas generator is available to provide electricity to the system. The power output of the biogas generator is given by 6 while the energy calculation is similar to solar energy calculation based on the power output shown in 5.

$$P_{out} = \frac{Q * CV * \eta}{t} \quad (6)$$

Where P_{out} is maximum power output of the biogas generator (in kW), Q is the flow rate of biogas in cubic meters per hour (m^3/h), CV is the calorific value of the biogas (in kJ/m^3), η is the system efficiency including the generator efficiency and the gas conversion efficiency, while t is the period of observation. The levelized cost of electricity (LCOE) offered by the biogas generator embeds the 15-year lifespan of the setup in the operational cost of the model.

The dataset is pre-processed in terms of data cleaning, feature selection, correlation, and inter/intra mapping to identify the following aspects of the data:

- Identify the period of next EV arrival along with energy requirement.
- Identify the high/low electricity price periods along with availability of the grid.
- Identify the period of solar energy generation.
- Identify the biogas generator availability period.
- Interlink the identified periods as an input to the Optimization Model.

4. Proposed Methodology

The proposed architecture of the battery swapping station optimization model is shown in Figure 1. The battery swapping station (BSS) has two modes of operation comprising the swapping/replacement of a fully charged battery (FCB) with a depleted battery (DB) of the arrived EV, and the orderly charging of DB to meet the incoming battery demand. The BSS infrastructure consists of a centralized control center, multiple charging bays, battery stock, and a fully charged battery inventory (FCBI). In addition to the assets the BSS infrastructure also includes a charging network which is connected to the available generator and a battery management system (BMS) to monitor the power flows and system health.

The two modes of operation are reflect on the daily finances of the BSS such that the battery swap/replacement generates revenue for the BSS while the charging of the DB results in the cost incurred on the BSS operation. The daily profit or loss depends on the revenue and cost, i.e., $Profit/Loss = Revenue - Cost$. The problem to enhance the daily profit of the BSS is a two-fold problem where the daily revenue needs to be increased while reducing the daily incurred cost.

To achieve this goal of reduced cost, the proposed architecture introduces a generation mix comprising of grid electricity, solar PV electricity, and biogas electricity. The generation mix operate in an optimal manner considering the peak and off-peak load demand along with electricity prices to ensure the cost of charging at the BSS is reduced.

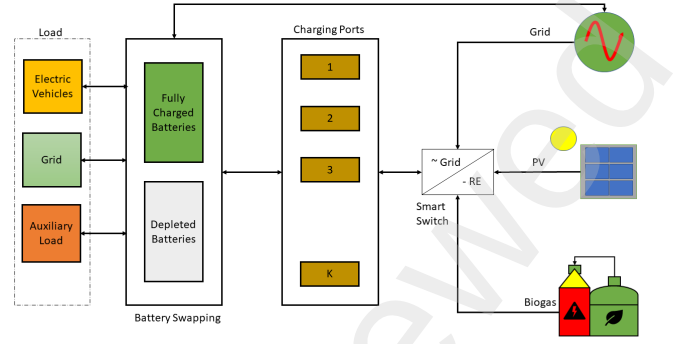


Figure 1: Proposed BSS Architecture

The introduction of solar PV and biogas energy into the generation mix allows the utilization of green energy with reduced emission in addition to the overall reduced cost of electricity. Moreover, to increase the revenue, the proposed architecture introduces the concept of discharging the available battery energy to the grid, i.e., BSS2G, especially during peak-hours such that the normal BSS operation, i.e., EV battery swapping, is not affected.

The approach aims to efficiently utilize the available generation and energy storage component of batteries available at the BSS. The flowchart of proposed methodology is shown in Figure 2. The proposed architecture first collect data from the connected grid, solar PV panels, and biogas plant. The data is feed into a hybrid energy mix along with the EV energy demand. If the available generation is less than the energy demand, the system performs load shedding (demand cut-off) to reduce demand such that the generation is greater than or equal to demand. When the available generation is greater than or equal to the demand the generation forecast and expected demand is provided to the Generation Optimization Model.

If the Generation Optimization Model generates feasible solution, the forwarding process generates the optimal energy price spread over the particular time horizon. This optimal price along with the bilateral energy trade with connected and arriving EV consumers, is provided to the BSS Optimization Model. The feasible solution of the BSS Optimization Model generates the expected energy demand output which is feed-back into the Generation Optimization Model along with the Net Daily Profits. The resultant output of the research problem is the net daily profit which is the main evaluation metrics.

To ensure the incorporation of batteries placed at the BSS in the BSS Optimization Model, each battery is associated with a few characteristic including a unique id, current SOC, energy available, and current state of the battery. The battery state can be any from the state set of four states given as $States = \{1 : Fully - Charged, 2 : Swapped, 3 : Charging, 4 : Discharging\}$. A battery can exists in only one state at a time and the state transition diagram of the battery at the BSS is shown in Figure 3.

Moreover, to avoid undesirable scenarios, the model managed and controls both charging and discharging rates based on the price signals, energy demand, and battery energy storage component of the BSS. In addition, to the monitoring and control the model includes the battery degradation considering the charging and discharging cycles. The model is simulated on three different scenarios in Generation Optimization Model and six different scenarios in BSS Optimization Model to analyze the profit earnings and performance evaluation. The generation scenarios include the utilization of grid only mode, grid and solar PV hybrid mode and grid, solar PV, and biogas hybrid mode. The BSS scenario considers the three generation modes and two scenarios with and without discharging to grid is allowed. Based on this proposed architecture the next section described the mathematical formulation of the model.

5. Mathematical Formulation

The BSS serves as the main hub and control center of the architecture, the EV arrival and SOC forecast is given as an input to the BSS and the BSS provides the expected energy forecast in addition to the auxiliary demand of the station as output.

Tool: The model is implemented and simulated in MATLAB, including data pre-processing, constraints, model formulation, and deduction of graphical and meaningful results. The Generation Optimization Model is formulated to minimize the cost of electricity as a Mixed Integer Programming (MIP) and linearized to reduced complexity as Mixed Linear Integer Programming (MILP) while the BSS Optimization is formulated to maximize the daily profit as time-dependant dynamic programming problem given by 7.

$$\max f = \max \{R - C\} \quad (7)$$

Where f is the objective function, i.e., profit/loss obtained as difference of revenue R and cost C . In the proposed architecture the BSS can generate revenue by each swapping batteries with EVs or by discharging energy storage to the grid as given by 8.

$$R = R^{swap} + R^{dis} \quad (8)$$

Where R^{swap} is the revenue generated from swapping battery and R^{dis} is the revenue generated from discharging the energy stored back to grid. The revenue generated from swapping battery consists of two components, i.e., fix cost of swapping applied to each battery swapping process and the cost of energy being swapping as given by 9.

$$R^{swap} = R^{swap-fix-cost} + R^{swap-energy-traded} \quad (9)$$

Where $R^{swap-fix-cost}$ is the fix revenue while the variable $R^{swap-energy-traded}$ is the revenue generated by trading energy in the process of swapping. The fix cost and energy trade revenue is given by 10 and 11 respectively.

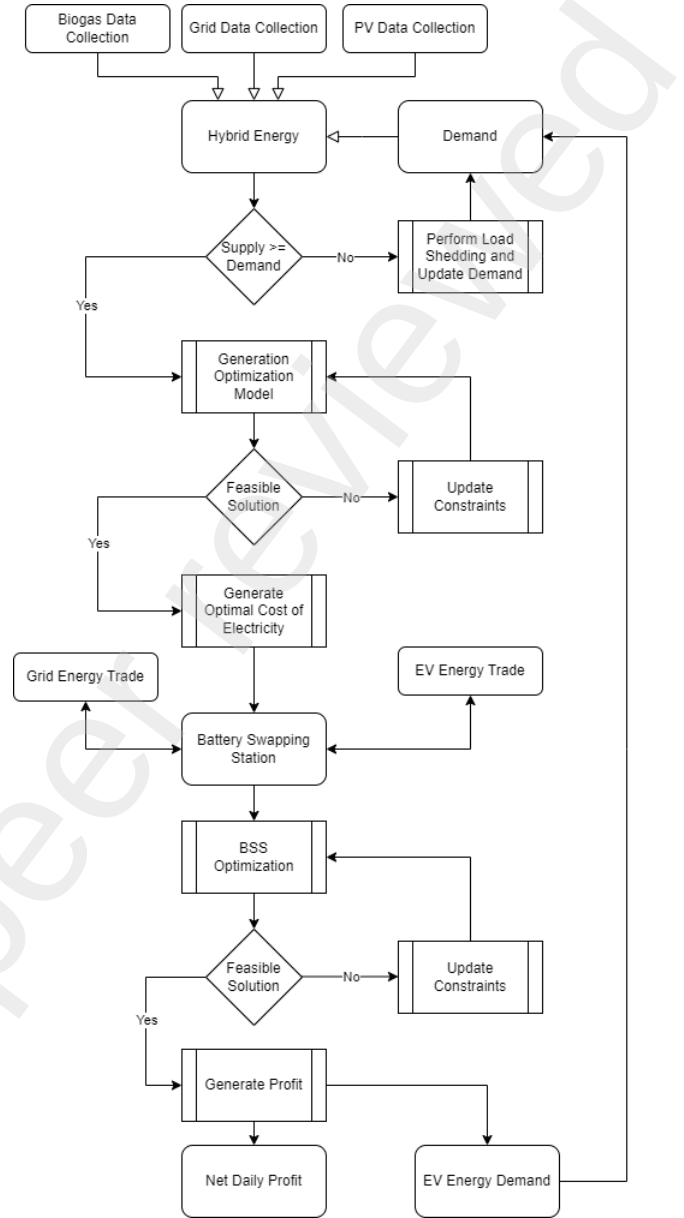


Figure 2: Flowchart of Proposed Methodology

$$R^{swap-fix-cost} = \sum_{t=1}^T \sum_{b=1}^B swap_{t,b} * FC_{swap} \quad (10)$$

$$R^{swap-energy-traded} = \sum_{t=1}^T \sum_{b=1}^B Q_b * E_{t,b}^{BSS2EV} * P_t^{BSS2EV} \quad (11)$$

Where $swap_{t,b}$ is the binary variable which identifies whether or not battery b is swapped at period t given by 12, FC_{swap} is the fix cost associated with each swap, Q_b is the battery b capacity (in kWh), $E_{t,b}^{BSS2EV}$ is the energy traded from the BSS to EV (BSS2EV) for battery b swapped at time

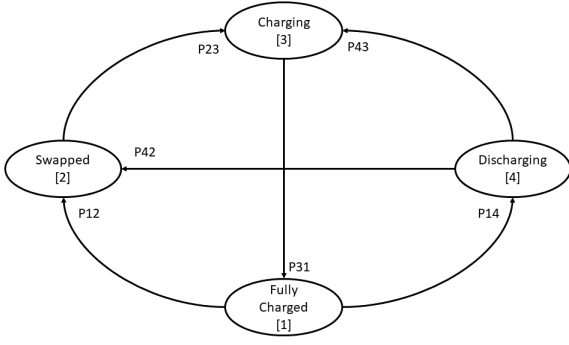


Figure 3: State Diagram of Battery at BSS

t , and P_t^{BSS2EV} is the price of electricity offered by BSS to EV at time t .

$$swap_{t,b} = \begin{cases} 1 & \text{battery } b \text{ swapped at } t \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

The revenue generated by the BSS by discharging excess energy storage available is given by 13. In addition to revenue, whenever the BSS batteries are allowed to discharge and sell energy to the grid, there is an associated cost of discharging which highlights the battery degradation cost described in the following formulation.

$$R^{dis} = \sum_{t=1}^T \sum_{b=1}^B P_t^g * E'_{t,b}^{dis} * \eta^{dis} \quad (13)$$

Where P_t^g is the electricity price related to the connected grid consisting of peak and off-peak hours, $E'_{t,b}^{dis}$ is the change in energy observed by the BSS when discharging is carried out by battery b at period t , while η^{dis} is the discharging efficiency associated with the inverter and cable loss in the connection path.

In the proposed architecture the BSS cost is related to the cost of charging batteries denoted by C^{chg} , the charging cycle degradation cost $C^{fix-cost-chg}$, and the discharging cycle degradation cost $C^{fix-cost-dis}$ given by 14.

$$C = C^{chg} + C^{fix-cost-chg} + C^{fix-cost-dis} \quad (14)$$

$$C^{chg} = \sum_{t=1}^T \sum_{b=1}^B E'_{t,b}^{chg} * \eta^{chg} * P_t^{opt} \quad (15)$$

$$P_t^{opt} = \min \sum_{t=1}^T x_t^g * w_t^g * g_t^{avl} + x^p v_t * w^p v_t + x^b g_t * w^b g_t * b g_t^{avl} \quad (16)$$

Where $E'_{t,b}^{chg}$ is the difference in energy observed at battery b at period t when battery is placed on charging, η^{chg} is the charging efficiency associated with the converter and cable loss, and P_t^{opt} is the optimal price generated by the Generation Optimization Model. The P_t^{opt} is the minimization for cost of electricity given by 16 where x_t^g denotes the number of units sold by the grid at period t , w_t^g is the cost of electricity offered by the grid at time t , and g_t^{avl} is the grid availability whether or not grid is available or not given by 17.

$$g_t^{avl} = \begin{cases} 1 & \text{grid is available at } t \\ 0 & \text{otherwise} \end{cases} \quad (17)$$

$$b g_t^{avl} = \begin{cases} 1 & \text{biogas generator available at } t \\ 0 & \text{otherwise} \end{cases} \quad (18)$$

Similarly $x^p v_t$ is the number of units sold by the solar PV at time t and $w^p v_t$ is the price of electricity offered by the solar PV panels at time t . Similarly $x^b g_t$ is the number of units sold by the biogas generator at time t , $w^b g_t$ is the price of electricity offered by the biogas generator at time t , and $b g_t^{avl}$ is the availability of biogas generator given by 18.

The cost of charging and discharging cycles is included in the optimization model which is determined based on the fix cost associated with charging and discharging given by 19 and 20 respectively.

$$C_t^{fix-chg} = \sum_{t=1}^T FC^{chg} * N_t^{chg} * chg_{t,b} \quad (19)$$

$$C_t^{fix-dis} = \sum_{t=1}^T FC^{dis} * N_t^{dis} * dis_{t,b} \quad (20)$$

Where FC^{chg} is the fix cost associated with charging cycles at time t , N_t^{chg} is the number of batteries fully charged at time t , and $chg_{t,b}$ is the battery state given by 21. Similarly, FC^{dis} is the fix cost associated with discharging cycles at time t , N_t^{dis} is the number of batteries being discharged at time t , and $dis_{t,b}$ is the battery state given by 22.

$$chg_{t,b} = \begin{cases} 1 & \text{battery } b \text{ is being charged at } t \\ 0 & \text{otherwise} \end{cases} \quad (21)$$

$$dis_{t,b} = \begin{cases} 1 & \text{battery } b \text{ is being discharged at } t \\ 0 & \text{otherwise} \end{cases} \quad (22)$$

Based on the revenue and cost described in 8 to 22, the objective function 7 is updated in two methods, the first one does not allow discharging to grid given by 23.

$$\begin{aligned}
 f = & \sum_{t=1}^T \sum_{b=1}^B \text{swap}_{t,b} * FC_{\text{swap}} \\
 & + \sum_{t=1}^T \sum_{b=1}^B Q_b * E_{t,b}^{BSS2EV} * P_t^{BSS2EV} \\
 & - \sum_{t=1}^T \sum_{b=1}^B E_{t,b}'^{chg} * \eta^{chg} * P_t^{opt} \\
 & - \sum_{t=1}^T FC^{chg} * N_t^{chg} * chg_{t,b}
 \end{aligned} \quad (23)$$

While the second method allows discharging to the grid given by 24, whereas the parameter P_t^{opt} is an optimization problem defined as follows.

$$\begin{aligned}
 f = & \sum_{t=1}^T \sum_{b=1}^B \text{swap}_{t,b} * FC_{\text{swap}} \\
 & + \sum_{t=1}^T \sum_{b=1}^B Q_b * E_{t,b}^{BSS2EV} * P_t^{BSS2EV} \\
 & + \sum_{t=1}^T \sum_{b=1}^B P_t^g * E_{t,b}'^{dis} * \eta^{dis} \\
 & - \sum_{t=1}^T \sum_{b=1}^B E_{t,b}'^{chg} * \eta^{chg} * P_t^{opt} \\
 & - \sum_{t=1}^T FC^{chg} * N_t^{chg} * chg_{t,b} \\
 & - \sum_{t=1}^T FC^{dis} * N_t^{dis} * dis_{t,b}
 \end{aligned} \quad (24)$$

The Generation Optimization Model's objective function given by 16 is defined as follows.

$$\begin{aligned}
 P_t^{opt} = & \min \sum_{t=1}^T x_t^g * w_t^g * g_t^{avl} + x^p v_t * w^p v_t + x^b g_t * \\
 & w^b g_t * b g_t^{avl} \\
 \text{subject to:}
 \end{aligned}$$

$$x_t^g + x^p v_t + x^b g_t = D_t \quad (25)$$

$$x_t^g \leq G_{\max} \quad (26)$$

$$x^p v_t \leq PV_{\max} \quad (27)$$

$$x^b g_t \leq BG_{\max} \quad (28)$$

$$\{x^g, x^p v, x^b g\} \geq 0 \quad \forall t \quad (29)$$

$$\{g^{avl}, b g^{avl}\} \in \{0, 1\} \quad \forall t \quad (30)$$

Where $G_{\max}$ is the maximum grid capacity, $PV_{\max}$ is the maximum PV panel capacity, $BG_{\max}$ is the maximum biogas generator capacity, and D_t is the energy demand at time t . The BSS Optimization Model's objective function given by 23 and 24 is defined as follows.

Objective function:

$$\begin{aligned}
 f = & \sum_{t=1}^T \sum_{b=1}^B \text{swap}_{t,b} * FC_{\text{swap}} \\
 & + \sum_{t=1}^T \sum_{b=1}^B Q_b * E_{t,b}^{BSS2EV} * P_t^{BSS2EV} \\
 & + \sum_{t=1}^T \sum_{b=1}^B P_t^g * E_{t,b}'^{dis} * \eta^{dis} \\
 & - \sum_{t=1}^T \sum_{b=1}^B E_{t,b}'^{chg} * \eta^{chg} * P_t^{opt} \\
 & - \sum_{t=1}^T FC^{chg} * N_t^{chg} * chg_{t,b} \\
 & - \sum_{t=1}^T FC^{dis} * N_t^{dis} * dis_{t,b}
 \end{aligned}$$

subject to:

$$E_t^{BSS2EV} = \begin{cases} (E_{t-1}^{BSS} - E_t^{EV} * \text{swap}_{t,b}) & \forall t > 1 \\ (E_t^{BSS} - E_t^{EV} * \text{swap}_{t,b}) & t = 1 \end{cases} \quad (31)$$

Where 31 performs the energy updates once a battery swapping operation occurs.

$$E_{t,b}'^{dis} = \begin{cases} -\frac{dE_{t,b}^{BSS}}{dt} & E_{t,b}^{BSS} > \gamma^{\min\text{-energy-dis}} * Q_b \\ 0 & \text{otherwise} \end{cases} \quad (32)$$

$$-\frac{dE_{t,b}^{BSS}}{dt} = \begin{cases} E_{t-1}^{BSS} - E_t^{BSS} * dis_{t,b} & \forall t > 1 \\ 0 & t = 1 \end{cases} \quad (33)$$

$$E_{t,b}'^{chg} = \begin{cases} \frac{dE_{t,b}^{BSS}}{dt} & \gamma^{\min\text{-energy-chg}} * Q_b \\ 0 & t = 1 \end{cases} \quad (34)$$

$$\frac{dE_{t,b}^{BSS}}{dt} = \begin{cases} E_t^{BSS} - E_{t-1}^{BSS} * chg_{t,b} & \forall t > 1 \\ 0 & t = 1 \end{cases} \quad (35)$$

Where 32 and 34 associate the energy charge and discharge of battery b at time t , while $\gamma^{\min\text{-energy-dis}}$ and $\gamma^{\min\text{-energy-chg}}$ denotes the percentage of beyond which energy discharging is not allowed and battery charging starts respectively.

$$E_{t,b}^{BSS} = \begin{cases} (E_t^{EV} * \text{swap}_{t,b}) + \\ (E_{t-1,b}^{BSS} + (E_{t,b}'^{dis} * dis_{t,b}) \\ + (E_{t,b}'^{chg} * chg_{t,b})) & \forall t > 1 \\ E_{t,b}^{BSS} & t = 1 \end{cases} \quad (36)$$

Where 36 updates the associated energy of BSS battery b at time t .

$$swap_{t,b} + chg_{t,b} + dis_{t,b} \leq 1 \quad (37)$$

Where 37 reflect at battery b can only be swapped or placed at charging or being charged at time t .

$$E_{min}^{BSS} \leq E_{t,b}^{BSS} \leq E_{max}^{BSS} \quad (38)$$

Where 38 limits the maximum and minimum energy of battery b at time t .

$$\sum_{b=1}^B chg_{t,b} \leq N^{chg-ports} \quad (39)$$

Where 39 states that the number of batteries placed on charging are limited the number of charging ports $N^{chg-port}$ available.

$$E_{t-1}^{BSS} \geq \gamma^{PCB} * Q_b * swap_{t,b} \quad (40)$$

Where 40 represents that the batteries at BSS can be swapped even if they are partially charged where the threshold beyond which the battery is accepted is controlled in terms of percentage by γ^{PCB} .

$$E'_{t,b}^{dis} \leq \gamma^{dis} * Q_b * dis_{t,b} \quad (41)$$

$$E'_{t,b}^{chg} \leq \gamma^{chg} * Q_b * chg_{t,b} \quad (42)$$

Where 41 and 42 represents that the energy discharge and charge from the batteries at is limited and controlled in terms of percentage by γ^{dis} and γ^{chg} respectively.

$$E'_{t,b}^{dis} \in T' \quad (43)$$

$$T' = \{t | t \in T \& N_t^{EV} \leq \gamma^{max-dis-limit} * N^{Battery}\} \quad (44)$$

Where 43 and 44 state that the battery discharged is allowed when the number of EV arrived is less than a threshold value given in percentage by $\gamma^{max-dis-limit}$.

$$\sum_{b=1}^B dist_{t,b} \leq \gamma^{allowed-dis} * N^{Battery} \quad (45)$$

$$\{swap_{t,b}, chg_{t,b}, dis_{t,b}\} \in \{0, 1\} \quad \forall t \quad (46)$$

Where 45 state that the number of batteries placed on charged are limited by a percentage given by $\gamma^{allowed-dis}$ and 46 states that these variable are binary variables. The next section describes the experimental setup as per the ongoing research pilot project.



Figure 4: Battery Swapping Station installed at the ongoing project

6. Experimental Setup

The mathematical formulation devised is based on the ongoing research pilot project which is focused on the promotion and adaptability of two and three wheeler Electric Vehicles in rural areas, especially areas where either no grid is available or the connected grid is either weak/unreliable. A similar pilot research project is currently ongoing with two battery swapping stations currently operating to provide fully charged batteries to hundred two and three Wheeler's EV in our designated rural area as shown Figure 4.

Each BSS has a total of sixteen (16) batteries in stock and one battery is kept as spare along with five (05) charging ports. Therefore, we set the total number of batteries available $N^{Battery} = 30$ and number of charging ports $N^{chg-ports} = 10$. Each battery has an energy capacity of 5kWh (or 100Ah ampacity), i.e., $Q_b = 5$. The auxiliary demand of the both BSS is 10kW per hour, i.e., $D_t^{aux} = 10$ and the period of daily operation is a 24 hour duration at a granularity of 1 hour time period.

At the swap the BSS charges a fix cost of 100/swap of swapping operation $FC^{swap} = 100$, and the per unit energy traded with the EV is done at a unit price of 60, i.e., $P_t^{BSS2EV} = 60/kWh$ which is obtained as the weighted average cost of generation (WACG). The battery charging rate is a controllable parameter whose value is typically kept around 0.5C $\gamma^{chg} = 0.5$, similarly, for our use case we have introduced a controllable discharging rate and it is kept at a low value of 0.1C, i.e., $\gamma^{dis} = 0.1$.

The battery charging process starts if a battery energy drops below a certain threshold typically kept high as close to 90%, however, in our use case to avoid redundant charging and discharging cycles the battery charging is carried out in an orderly manner such that the lower limit is kept at 60% $\gamma^{min-energy-chg} = 0.6$. Similarly, the limit beyond which battery discharging is not allowed is also set to 60% $\gamma^{min-energy-dis} = 0.6$. As we consider that batteries which are not fully charged can also be provided for swapping given they have sufficient energy beyond the threshold of 70% SOC $\gamma^{PCB} = 0.7$.

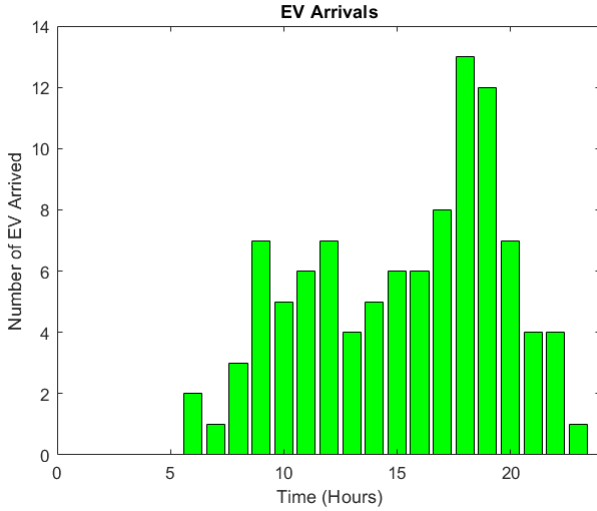


Figure 5: Number of EVs Arriving at the BSS

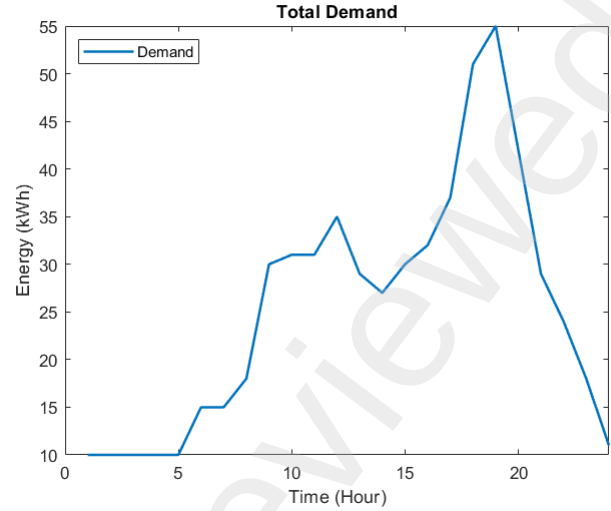


Figure 6: Energy Demand when Discharging is Not Allowed

Moreover, the battery discharging process can only be allowed in time period T' where the number of EVs arriving at the BSS are less than the user-defined limit given by $\gamma^{max-dis-limit} = 0.3$ such that discharging will continue until the number of EV arriving is more the 30% of the capacity. In addition to the discharging period a maximum of 50% of the batteries are allowed to be discharged which is also user-defined parameter given by $\gamma^{allowed-dis} = 0.5$. The experimental setup is simulated on six different scenarios described in Table 1.

 Table 1
Simulation Scenarios

Scenario	Available Generation	BSS-To-Grid
Scenario-1	Grid	Not allowed
Scenario-2	Grid	Allowed
Scenario-3	Grid and PV	Not allowed
Scenario-4	Grid and PV	Allowed
Scenario-5	Grid, PV, and biogas	Not allowed
Scenario-6	Grid, PV, and biogas	Allowed

Based on the experimental setup and simulation scenarios, the next section discuss the model output and results.

7. Results and Discussion

As discussed in the experimental setup, the base case demand occurs when we do not allow batteries to discharge and feed back to grid while the second scenario allows the discharging of batteries hence the overall demand changes. For a given day the EV arrival forecast is predicted as shown in Figure 5 a maximum of 13 EV consumers arriving in a given hour to swap their depleted batteries. The energy demand as observed by the BSS based on the expected EV arrival forecast is shown in Figure 6.

If the EV Arrival forecast is provided as an input to the BSS Optimization Model and the model is allowed

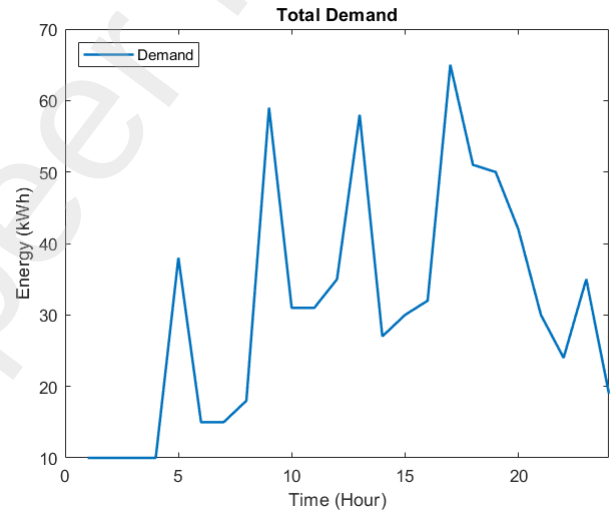


Figure 7: Energy Demand when Discharging is Allowed

to discharge the available idle batteries to the grid while operating within the constraints, the model generates a secondary energy demand profile. This demand incorporates EV battery swapping energy requirement in addition to the energy required by the BSS to fulfill the energy depleted from the discharged batteries the BSS idle. The two demand profiles vary significantly and play a crucial role in financial calculations. The demand when idle BSS batteries are allowed to be discharged based on the forecast is shown in Figure 7. Both demand profiles when given as input to the Generation Optimization Model, generate the optimal energy price required by the BSS Optimization Model and determine the minimum cost of electricity. The detail of units sold by the grid, solar PV, and biogas generator after optimization in the secondary demand profile with allowed discharging is shown in Figure 8.

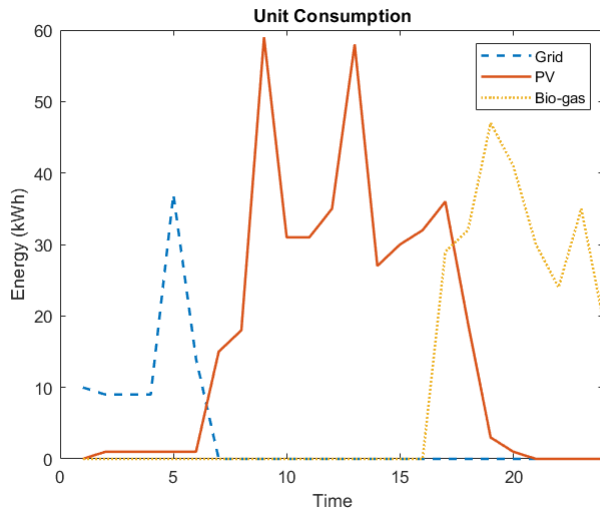


Figure 8: Optimized Unit Consumption Detail for Demand with Allowed Discharging

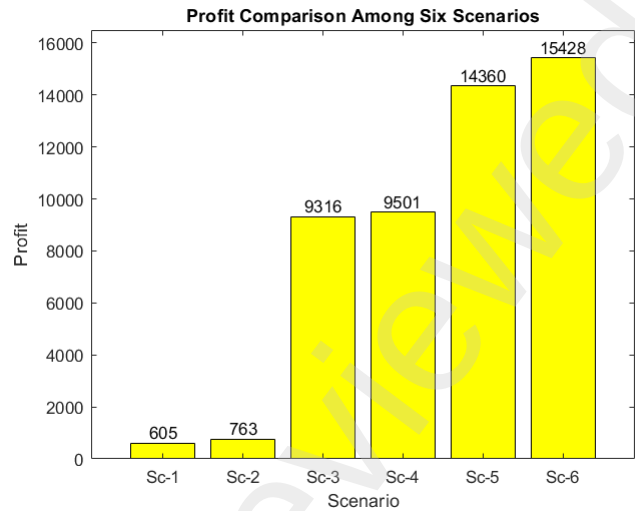


Figure 10: Profit Comparison of Six Scenarios

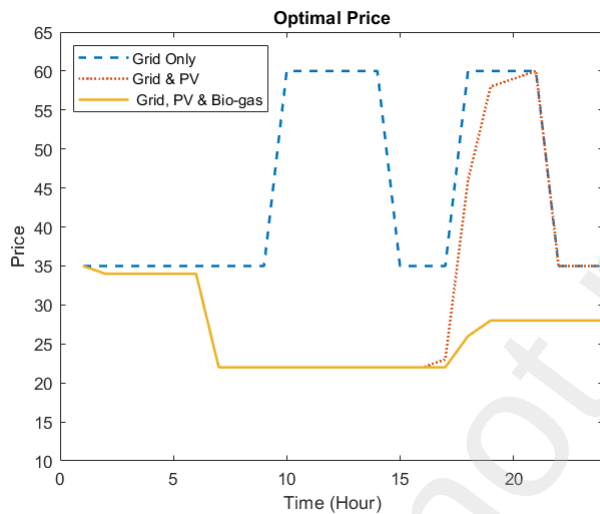


Figure 9: Optimal Price of Electricity in Different Generation Mix Scenarios

From Figure 8, it is evident that the Generation Optimization Model tries to minimize the cost of electricity by maximally utilizing the cheapest generation resource, i.e., Solar PV, whenever it is available, followed by the next cheapest resource, i.e., biogas generation, while keeping the grid energy as a last resort. Considering the simulation scenarios, where different generation resources are available such as grid only, grid and PV, and grid, PV and biogas, each scenario generates an optimal price as an output of Generation Optimization Model in addition to the unit consumption detail. This optimal price is required as an input to the BSS Optimization Model and it is shown in Figure 9.

The Figure 9 it is evident that the using the generation mix if grid, PV, and biogas will drastically reduce the overall cost of battery charging. Moreover, the six scenarios tabulated in Table 1 takes optimal values from the same data and

generate revenue, cost and profit as shown in Figure 10 and it is observed that the difference between the base case and the fully function Optimization model is significant with a rise of revenue by 18%, reduction in cost by 33% and an overall increase in net profit by almost 20 times as compared to the base case. Moreover, an increase of almost 8% is observed in Scenario-6 with Grid/PV/Biogas generation with and without battery-to-grid mode. The revenue, cost and profit calculation are tabulated in Table 2.

Table 2

Financial Values of Simulated Scenarios over 24 hour Period

Scenario	Revenue	Cost	Profit
Scenario-1	29580	28975	605
Scenario-2	34850	34087	763
Scenario-3	29580	20264	9316
Scenario-4	34850	25349	9501
Scenario-5	29580	15220	14360
Scenario-6	34850	19422	15428

The profit shown in Table 2 is in Pakistani Rupee (PKR) which is equivalent to 0.0036 US Dollar, therefore, the profit earning in the table ranges between \$ 2.15 - \$ 55.50 US Dollar from Scenario - 1 to Scenario - 6. Based on the simulation results and profit comparison, it is evident that the proposed architecture significantly improves the daily profit of the BSS. However, the scenarios have been tested on a single day BSS operation which is a mid-week day. Therefore, we perform the same analysis for two more days of the week, i.e., Monday, and Saturday, such that the system observe different EV arrival forecast and energy requirement. The model is simulated for all six scenarios for each day and the results are summarized as follows.

Figure 11 shows the profit comparison across multiple days of the week, and the pattern previously discussed prevails, i.e., the use of proposed architecture significantly

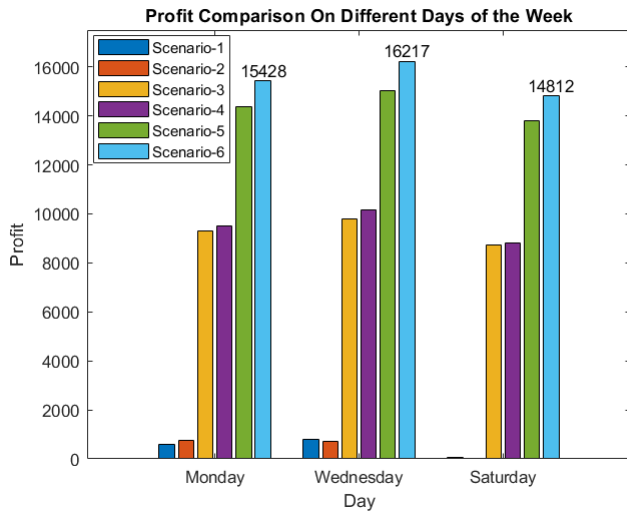


Figure 11: Profit Comparison across Days of the Week

increasing the daily net profit. Moreover, it is crucial to note that the Scenario-2 where the idle BSS batteries are allowed to discharge while charging the batteries from grid only source results in either generate loss or has reduced profit. This is due to the reason that the discharged batteries are again being charged from the same grid at the same rate while incorporating the battery charging and discharging degradation cost hence Scenario-2 is not recommended. While Scenario-6 which allows batteries to be discharged along with the utilization of grid, PV, and biogas generation mix generates highest profit, the Scenario-6 is highly recommended to be opted for the enhancement of BSS daily net profit.

It is pertinent to mention that whenever the BSS received a swapping request from the EV, the optimization model prioritize the request and tries to serve the EV swapping request before any other optimization operation. In addition to the priority, the model also keep record and tries to minimize the number of EV which have not been served due to lack of fully or partially charged batteries.

For the simulation the discharging of batteries is allowed when the number of EV arrival in a time period is less than 30%, this limit is obtained from the sensitivity analysis and is kept below the knee point as shown in Figure 12. It is observed that the profit increasing till the knee point and then it saturates identifying the BSS prioritize the swapping request over the battery discharging process. Since, the profit values of Saturday starts saturating around 40%, therefore, the simulation value is kept at 30% such that the model can incorporate the error in EV arrival forecast. Furthermore, the profit earning is maximum on Monday followed by Wednesday and Saturday being the least profitable day of the week.

From the results and discussion on the proposed architecture and simulation scenarios, a significant reduction of 18% in BSS operational cost as compared to the base case is observed due to the optimal generation mix comprising of Grid

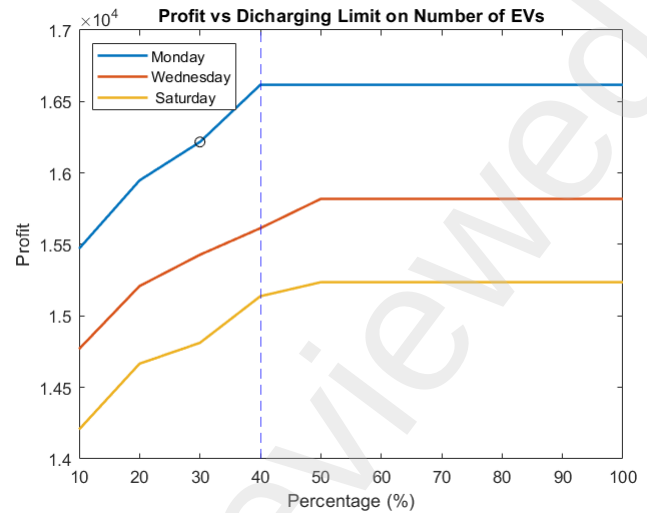


Figure 12: Profit Vs Discharging Limit based on Number of EV Arrivals

energy, Solar PV energy, and biogas energy. Furthermore, an increase the BSS revenue of 34% as compared to the base case is observed due to the discharging of idle batteries to the grid. The daily net profit of the BSS is increased by almost 20 times as compared to the base case scenario and by almost 8% when BSS is allowed to operate in battery-to-grid mode with discharging. The proposed architecture has potential to improve the adaptability and profitability of the BSS and the architecture can be further improved to feed other battery-operated appliances in addition to EV as a way to increase the number of revenue streams.

8. Conclusion

The battery swapping station address EV range anxiety and long charging time by offering quick energy replenishment by simply swapping the EV's depleted battery with a fully charged battery. However, the adoption of BSS architecture is hindered due to lack of infrastructure and profit earnings. Therefore, to promote the use of BSS architecture in EV energy replenishment, the proposed model explores the problem in two directions, i.e., increasing revenue and decreasing cost. The model demonstrates that the total revenue of the BSS increased by 18%, the total cost of the BSS is reduced by 34%, and the daily net profit increases by almost 20 times as compared to the base case or traditional BSS operation. Future improvement of the model include multiple other types of load which are battery operated and can work on the model of battery swapping. Such an inclusion will improve the battery utilization while reducing the battery deck time hence increasing the BSS operational profit. The enhanced model will be a crucial part of the ongoing research pilot project focused on the integration of energy, food and transportation (EFTN) sector.

CRedit authorship contribution statement

Muhammad Shuzub Gull: Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing - Original draft preparation. **Muhammad Khalid:** Investigation, Resources, Supervision, Validation, Visualization, Writing - review & editing. **Naveed Arshad:** Conceptualization, Resources, Supervision, Visualization, Writing - review & editing.

Declarations

The authors declare that they have no known competing financial interest or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be available on request.

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